Outlier Migration in Long Decode Traces

Anonymous authors
Paper under double-blind review

Abstract

This draft reports the first two preregistered gates for OutlierMigrate, a candidate positive-method branch studying whether high-magnitude activation channels remain stationary during long decode. Phase 0 passes on Granite-4.0-H-Tiny: the fraction of top-1% position-100 channels whose rank changes by more than two channel positions by decode position 10000 is 0.8178385416666667 with bootstrap 95% CI [0.797265625, 0.8368489583333334]. Phase 1 replicates the dynamic-outlier direction at larger scale on Granite-4.0-H-Small: the migration fraction is 0.843165650406504 with bootstrap 95% CI [0.8334349593495936, 0.8511432926829268]. This is a strong same-family observational gate for a positive-method branch, but it is not camera-ready and does not yet support cross-model or intervention claims. Cross-model validation and audit checks remain required before submission-level claims.

1 Introduction

OutlierMigrate asks whether the channels carrying high-magnitude decode-time activations are fixed or whether they move during long reasoning traces. Static outliers would support static channel-protection recipes. Dynamic outliers would motivate migration-aware protection or routing methods, but only after validation shows that the effect is stable, useful, and not specific to one model family.

The current paper status is deliberately limited. Phase 0 is a screen on ibm-granite/granite-4.0-h-tiny with 12 deterministic AIME-2025 reasoning traces. Phase 1 scales the same dynamic path to ibm-granite/granite-4.0-h-small with 24 deterministic traces and an extended decode-position grid. Together they support a replicated same-family dynamic-outlier observation under the preregistered metric. They do not support a camera-ready method claim, an end-to-end efficiency claim, or any cross-model claim.

2 Related Work

Mamba quantization and dynamic outliers. OutlierMigrate is not the first work to observe dynamic outlier behavior in Mamba-style models. In vision state-space models, QMamba reports that hidden state sequences exhibit highly dynamic variations and introduces Temporal Group Quantization for hidden states (Li et al., 2025). OuroMamba makes the point even more directly for vision Mamba models: it identifies dynamic activation-outlier variation across time steps as a failure mode for static post-training quantization and uses lightweight dynamic outlier detection during inference (Ramachandran et al., 2025). These results are important prior art. They establish that dynamic outliers are already known in the vision-Mamba setting.

Language-Mamba quantization work has emphasized a different operating regime. Mamba-PTQ shows that recurrent LLMs exhibit activation outlier channels and frames outlier-aware quantization as a first step (Pierro and Abreu, 2024). Quamba proposes a static 8-bit per-tensor recipe that suppresses maximum input values to selective SSMs and uses a Hadamard transform for output activations (Chiang et al., 2024). MambaQuant uses variance-aligned rotations and reports heavy-tailed, channel-uneven distributions in Mamba-family models (Xu et al., 2025). Quamba-SE softens the treatment of activation outliers with multiple adaptive scales but still evaluates a per-value quantizer rather than a decode-time channel migration policy (Chen and Hemani, 2026). OutlierMigrate asks which regime applies to hybrid LLMs in long reasoning traces: the static or calibration-time protection regime common in language deployment recipes, or the dynamic regime already documented in vision Mamba.

Static protection in LLM deployment. Several influential LLM quantization systems protect or transform salient activation structure using calibration-time statistics or fixed transformations. SmoothQuant migrates activation difficulty into weights using offline activation statistics (Xiao et al., 2023); AWQ protects a small set of activation-salient weight channels (Lin et al., 2023); QuaRot removes hidden state outliers with rotations (Ashkboos et al., 2024); KVQuant isolates KV cache outliers per vector and uses per-channel key quantization (Hooper et al., 2024); and BlockDialect assigns block-wise number formats from a learned formatbook (Jang and Tambe, 2025). These systems are not claimed to be wrong by our current evidence. The implication is narrower: if the protected high-magnitude set changes substantially during long decode in hybrid LLMs, static protection maps may need migration-aware updates or validation on the specific recurrent/linear-attention architecture. Several of these systems already include adaptive components, such as KVQuant's per-vector dense-and-sparse handling or BlockDialect's online activation format selection; our current evidence motivates stress-testing those designs on hybrid long decode, not declaring them failed.

Channel-wise gated recurrence. Recent efficient-recurrence and linear-attention architectures make data-dependent channel-wise memory control a central design axis. Gated Linear Attention (GLA) formulates linear attention as recurrent matrix-valued state with data-dependent gating (Yang et al., 2023). Gated DeltaNet combines adaptive memory erasure with delta-rule updates and reports hybrids that mix Gated DeltaNet with attention or Mamba2 layers (Yang et al., 2024). Moonshot AI's October 2025 Kimi Linear report is a recent example in this line: its Kimi Delta Attention (KDA) is described as channel-wise gated linear attention that extends Gated DeltaNet with finer-grained gating, and the released Kimi Linear model is a layerwise hybrid of KDA and Multi-Head Latent Attention (Moonshot AI Kimi Team, 2025). The official Qwen3.6 model card likewise describes a hybrid layout that interleaves Gated DeltaNet blocks with gated attention (Qwen Team, 2026). These sources motivate OutlierMigrate as an empirical test of whether high-magnitude channels stay fixed during long decode in hybrid language models. They do not imply that Kimi Linear, Qwen3.6, GLA, or other unmeasured models exhibit the migration measured in this draft.

3 Method

Phase 0 follows the frozen preregistration at experimental/outlier_migrate/phase0/preregister_om_phase0.md. The model is ibm-granite/granite-4.0-h-tiny at HuggingFace snapshot commit 791e0d3d28c86e106c9b6e0b4cecdee0375b6124. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11, with prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json.

Phase 1 follows the frozen preregistration at experimental/outlier_migrate/phase1/preregister_om_phase1.md. The model is ibm-granite/granite-4.0-h-small at HuggingFace snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39. The prompt set contains 24 deterministically selected AIME-2025 traces, indices 0–23, with prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json.

For every transformer layer, the runner records per-channel activation magnitudes. Phase 0 records decode positions {100, 500, 1000, 5000, 10000}; Phase 1 records {100, 500, 1000, 5000, 10000, 20000}. Each gate selects the top 1% channels by mean magnitude at position

Quantity	Phase 0	Phase 1
Decision	dynamic-outlier pass	replicated-at-scale pass
Migration fraction	0.8178385416666667	0.843165650406504
Bootstrap 95% CI	[0.797265625, 0.8368489583333334]	[0.8334349593495936, 0.8511432926829268]
Preregistered threshold	≥0.05 and CI lower >0.05	≥0.05 and CI lower >0.05
Artifact complete	true	true
Model	ibm-granite/granite-4.0-h-tiny	ibm-granite/granite-4.0-h-small
Trace count	12	24
Layer count / hidden size	40 / 1536	40 / 4096
Decode positions	100, 500, 1000, 5000, 10000	100, 500, 1000, 5000, 10000, 20000
Bootstrap seed	20260508	20260508

Table 1: OutlierMigrate Phase 0 and Phase 1 gate results. Phase 0 values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json. Phase 1 values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json.

100 and computes the fraction whose rank at the final preregistered position differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers with equal layer weighting.

4 Experiments

The Phase 0 result packet is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z. The checker output is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json. The activation artifact SHA is sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687 from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json.

The Phase 1 result packet is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z. The checker output is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/artifact_check.json. The activation artifact SHA is sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1 from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The preregistered dynamic-outlier pass rule in both gates requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. Phase 1 uses the same dynamic path because Phase 0 returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION.

5 Results

The exact Phase 0 provenance values are: model snapshot commit 791e0d3d28c86e106c9b6e0b4cecdee0375b6124, prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd, and activation artifact SHA sha256:2783aa329d82e8f43ea4f342e52f

0a25fc283b0b8971880f2da093739474d687; these are recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/model_provenance.json, experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json, and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json. The exact Phase 1 provenance values are: model snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39, prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e, and activation artifact SHA sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1; these are recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/model_provenance.json, experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json, and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The Phase 0 checker returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION. Its exact artifact reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05. The Phase 1 checker returned PASS_OM_PHASE1_REPLICATED_AT_SCALE. Its exact artifact reason was that the point estimate 0.84316565 is at least 0.05 and the CI lower bound 0.83343496 is greater than 0.05.

6 Discussion

On this Granite rank-migration surface, the Phase 0 result weakens a fixed position-100 outlier-map interpretation: under the preregistered metric, most position-100 top-1% channels do not remain rank-stationary through position 10000. Phase 1 promotes the dynamic-outlier hypothesis within the Granite family because the effect direction replicated on a larger model, twice as many traces, and an extended decode-position grid. This is not a first-observation claim about Mamba broadly. QMamba and OuroMamba already establish dynamic hidden-state or activation-outlier behavior in vision Mamba. The new evidence here is that the dynamic regime also appears under a frozen metric in hybrid LLM long-reasoning traces.

This is the strongest OutlierMigrate evidence so far and keeps the branch alive as a positive-method candidate. It does not establish a positive method yet, and it does not establish that migration-aware intervention improves model quality, latency, memory, or robustness. The highest-priority next gate is cross-model validation with audit checks that separate a reusable migration mechanism from a same-family artifact.

The theoretical mechanism is plausible but not yet empirically validated outside Granite. In channel-wise gated recurrence or linear-attention layers, independent per-channel decay rates imply that channels retain state at different rates. Low-decay channels can accumulate magnitude over long decode, while high-decay channels shed magnitude. Under heterogeneous inputs and heterogeneous decay rates, the composition of the top 1% channels by magnitude should therefore change across decode positions rather than remain fixed to the position-100 set. This mechanism makes dynamic outlier migration a natural concern for static channel-protection schemes: a calibration-time set can be correct early in a trace and stale later in the same trace. Kimi Linear's KDA, Qwen3.6-style Gated DeltaNet blocks, and GLA-style gated recurrence provide architectural examples where channel-wise or fine-grained data-dependent gating is a first-class operation (Moonshot AI Kimi Team, 2025; Qwen Team, 2026; Yang et al., 2024; 2023). This paragraph is a mechanism argument only; this draft has not measured migration on Kimi Linear, Qwen3.6, or GLA variants.

7 Limitations

This draft is not camera-ready. It reports two positive gates in one model family. There is no cross-model validation, no strict cross-family falsification pair, no contamination audit, no independent seed repeat beyond the recorded bootstrap procedure, and no intervention showing that migration-aware handling improves an application metric. The result may also depend on the chosen decode positions, top-channel fraction, and rank-delta threshold, all of which were fixed before analysis and should not be retuned on these packets.

The empirical measurement here is on Granite Mamba-2-style SSM hybrids. Phase 2 cross-validation must extend the same logic to additional hybrid model families before the paper claims architectural generality. A Nemotron-3-only validation would be a partial cross-family check, not a complete substitute for broader validation on delta-rule linear-attention hybrids such as Qwen3.6 and Kimi Linear. Generalization to RWKV-7, GLA-style systems, and other efficient-recurrence variants is not established by these results.

8 Reproducibility Statement

The Phase 0 result packet is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z. It includes environment.json, model_provenance.json, prompt_manifest.json, command_metadata.json, random_seed.json, metrics.json, bootstrap_ci.json, checker_result.json, and artifact_check.json. The recorded Phase 0 command was:

experimental/shared/run_phase0_branch.py --branch outlier_migrate

The Phase 1 result packet is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z. It includes the same artifact classes and records the Granite-4.0-H-Small snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39. The recorded Phase 1 command was:

experimental/outlier_migrate/phase1/run_om_phase1.py

The checker decisions and exact numbers in this draft are taken from the two result directories above.

9 Ethics Statement

This is an activation-analysis systems study on benchmark prompts. The current artifact does not make deployment, user-data, safety, or capability-improvement claims. The main ethical risk is overclaiming same-family Phase 0 and Phase 1 evidence as a validated cross-model positive method; this draft explicitly rejects that interpretation until cross-model validation and audits pass.

References

Yinglong Li, Xiaoyu Liu, Jiacheng Li, Ruikang Xu, Yinda Chen, and Zhiwei Xiong. QMamba: Post-Training Quantization for Vision State Space Models. arXiv:2501.13624, January 2025. https://arxiv.org/abs/2501.13624.

Akshat Ramachandran, Mingyu Lee, Huan Xu, Souvik Kundu, and Tushar Krishna. OuroMamba: A Data-Free Quantization Framework for Vision Mamba. arXiv:2503.10959, ICCV 2025. https://arxiv.org/abs/2503.10959.

Alessandro Pierro and Steven Abreu. Mamba-PTQ: Outlier Channels in Recurrent Large Language Models. arXiv:2407.12397, 2024. https://arxiv.org/abs/2407.12397.

Hung-Yueh Chiang, Chi-Chih Chang, Natalia Frumkin, Kai-Chiang Wu, and Diana Marculescu. Quamba: A Post-Training Quantization Recipe for Selective State Space Models. arXiv:2410.13229, 2024. https://arxiv.org/abs/2410.13229.

Zukang Xu, Yuxuan Yue, Xing Hu, Zhihang Yuan, Zixu Jiang, Zhixuan Chen, Jiangyong Yu, Chen Xu, Sifan Zhou, and Dawei Yang. MambaQuant: Quantizing the Mamba Family with Variance Aligned Rotation Methods. arXiv:2501.13484, 2025. https://arxiv.org/abs/2501.13484.

Yizhi Chen and Ahmed Hemani. Late Breaking Results: Quamba-SE: Soft-edge Quantizer for Activations in State Space Models. arXiv:2601.09451, DATE Late Breaking Results 2026. https://arxiv.org/abs/2601.09451.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. arXiv:2211.10438, ICML 2023. https://arxiv.org/abs/2211.10438.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. arXiv:2306.00978, MLSys 2024. https://arxiv.org/abs/2306.00978.

Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. QuaRot: Outlier-Free 4-Bit Inference in Rotated LLMs. arXiv:2404.00456, 2024. https://arxiv.org/abs/2404.00456.

Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization. arXiv:2401.18079, NeurIPS 2024. https://arxiv.org/abs/2401.18079.

Wonsuk Jang and Thierry Tambe. BlockDialect: Block-wise Fine-grained Mixed Format Quantization for Energy-Efficient LLM Inference. arXiv:2501.01144, ICML 2025. https://arxiv.org/abs/2501.01144.

Qwen Team. Qwen3.6-35B-A3B model card. Hugging Face, 2026. https://huggingface.co/Qwen/Qwen3.6-35B-A3B.

Songlin Yang, Bailin Wang, Yikang Shen, Rameswar Panda, and Yoon Kim. Gated Linear Attention Transformers with Hardware-Efficient Training. arXiv:2312.06635, 2023. https://arxiv.org/abs/2312.06635.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated Delta Networks: Improving Mamba2 with Delta Rule. arXiv:2412.06464, 2024. https://arxiv.org/abs/2412.06464.

Moonshot AI Kimi Team: Yu Zhang et al. Kimi Linear: An Expressive, Efficient Attention Architecture. arXiv:2510.26692, October 2025. https://arxiv.org/abs/2510.26692.